

## Something to Crow About: What Avian Brains Can Tell Us About the Evolution of Intelligence

“In a spell of dry weather, when the Birds could find very little to drink, a thirsty Crow found a pitcher with a little water in it. But the pitcher was high and had a narrow neck, and no matter how he tried, the Crow could not reach the water. The poor thing felt as if he must die of thirst. Then an idea came to him. Picking up some small pebbles, he dropped them into the pitcher one by one. With each pebble the water rose a little higher until at last it was near enough so he could drink. Moral: In a pinch a good use of our wits may help us out.”

In Aesop’s fable, the Crow cleverly uses stones to raise the water level of the pitcher so that he can quench his thirst (Winter & Aesop, 1994). In fact, the intelligence of birds has appeared in myths across times and cultures. In Norse mythology, Odin had two wise ravens that would inform him of the news of the world at the end of each day (Hollander, 1995). In the myths of the indigenous peoples of the North American Pacific Northwest, the raven is the Creator of the world and a trickster god (Erdoes & Ortiz, 1984). Celtic mythology depicts crows as oracles, who could predict the fates of men in battle (Green, 1998). The question is, do these myths hold any water? How smart are birds? Can they use reasoning to solve problems? Can the ways that birds learn, think, and behave give us general insights into intelligence?

The topic of avian intelligence has only very recently entered into the purview of neuroscientific study. Psychologists have used pigeons to study learning and memory since the 1950s, but other species of birds were largely ignored. Although pigeons are excellent at visual discrimination and have relatively good memory, they are not widely considered to be exceedingly intelligent. It can take them hundreds of trials to learn tasks in the lab, and they aren’t particularly flexible when it comes to applying this kind of learning to new tasks. As Nathan Emery (2016) puts it, they’re not exactly “avian Einsteins.”

There are 10,000 species of birds, though, and pigeons are by no means representative of mental abilities across species. So, why have we only recently become interested in studying intelligence in other species of birds? One reason is that avian brains look very different from our own and were historically thought to function differently as well. Ludwig Edinger, the father of comparative neuroanatomy, argued that the higher-order areas in avian brains were primitive and only consisted of the basal ganglia, a brain region that “controls instinctive behavior” (Ebert, 2005). He asserted that the mental capacity of birds was limited, a viewpoint that persisted in neuroscience and hindered a closer examination of avian cognition until the end of the 20th century (Emery, 2016). Several studies during the ‘90s identified behaviors in birds that had previously been considered to be unique to apes, helping to thrust the field of avian cognition into the spotlight. In particular, psittacidae (parrots) and corvidae (the crow family) have been found to have mental abilities that can rival apes, our closest relatives. This is particularly remarkable when we consider that the two groups evolutionarily diverged over 300 million years ago. Evolution produced very different brains in these two groups, but they both seem to be able to solve similar kinds of problems. But do they process information in the same way? Comparing these two systems could help us understand better why the human brain is unique, and even why it might have evolved in the way that it did.

In some ways, the brains of birds and humans are similar. Certain groups of birds, like corvids and parrots, have a high amount of brain encephalization (brain to body ratio), just like

humans and apes. There is debate whether encephalization is the basis of intelligence, but it certainly is interesting that birds evolved relatively large brains given that brain tissue is so energetically expensive. And in birds specifically, who also have to use a large amount of their energy to power flight, having relatively large brains can almost seem evolutionary counterintuitive (Emery, 2016). There seems like there should be a strong selective pressure to devote as much energy as possible to flight, and to reduce their overall body weight (Taylor, 2014). So, that indicates that having such a large brain must have some kind of evolutionary advantage that outweighs its cost.

On the surface—literally—avian brains are structured quite differently from mammal brains. Avian brains notably don't have sulci and gyri, the grooves and ridges that are found on mammalian brains. In mammals, this folding of the cortex helps increase its surface area without having to increase the total volume of the brain. Even so, avian brains have been found to contain about twice as many neurons as equally-sized primate brains (Olkowicz et al., 2016). Interestingly, the way that these neurons are organized in each brain is also distinct. Avian brains have a nuclear organization, in contrast to the laminar organization of mammalian brains. The cortex of mammalian brains have distinct layers of neuron cell bodies, with connections that extend to other parts of the brain. Avian brains, on the other hand, have significantly shorter neural projections because their neurons are organized in clusters and their cell bodies are found throughout the brain (Emery 2016). But the organization of avian brains are not as different from mammalian brains as they might first appear. On some level, they have been found to be organized in the same ways and seem to have many of the same network properties (Shanahan et al., 2013). In both of these networks, each node is relatively closely connected, which is thought to enable efficient communication between them while also allowing for functional specialization (Shanahan et al., 2013). In fact, the shared network properties, rather than the high amount of brain encephalization, might be a good way to understand the cognitive similarities between birds and primates.

In this piece, I'll discuss aspects of avian cognition that can potentially give us insights into how intelligence can arise in different kinds of brain. When I use the word "cognition," I'm referring to information processing and internal representations in the brain, as well as the application of these representations to drive behavior. I will use "intelligence" to mean the ability to flexibly use cognitive abilities in novel situations. Cognition is a mechanism, something that is usually happening in the brain in response to an input. Intelligence is a behavioral output, not a mechanism. Certain kinds of birds and apes have similar behavioral outputs (intelligence) in response to given inputs, but do their brains process and encode the information in the same way? Considering how they are similar and, most importantly, how they are different, can give us helpful insights into how brain organization leads to cognitive information processing.

### Mental time travel

Birds, non-human primates, and humans are all animals that seem to have strong abilities to remember the past and plan for the future. The mental time travel (MTT) hypothesis refers to the ability to remember a personal past and to imagine a personal future, and is therefore strongly tied to our concept of self. The central idea of this hypothesis, though, is that these abilities are uniquely human. Due to its dependence on "self," it can be challenging to

disprove this hypothesis experimentally since we can't exactly ask animals what they are thinking, or if they have a sense of self. But we can test aspects of this hypothesis in other animals.

Many birds will hide, or cache, their food so that they can eat it later. After a certain time period, they will recover and eat their previously cached food. Some foods that birds cache degrade more quickly than others, so they need to be recovered within a shorter time period. In order to keep track of this, birds would have to have some kind of cognitive mechanism to store information relating to *what* food was stored, *where* it was hidden, and *when* it was put there. Can birds store information in this way, and can they use this information to make decisions about food recovery?

Episodic memory in humans is the memory of past personal experiences that are associated with a specific time and place. Episodic memory has a what-where-when component, but also requires a conscious sense of self. Because of this added consciousness component, cognitive scientists have been hesitant to say that non-human animals might have episodic memories. But studies do seem to indicate that Western scrub jays, a kind of corvid, do at least have what-where-when memories that have been described as "episodic-like" (Clayton & Dickinson, 1998).

To simulate caching and recovery in the lab, Clayton and Dickinson (1998) filled ice cube trays with sand and let the scrub jays use them to cache food. They split the birds into two groups: the Degrade Group or the Replenish Group. In the Degrade Group, jays cached worms first (their preferred food), waited 120 hours, then were given access to the trays again to cache peanuts. In the Replenish Group, birds first cached worms and then peanuts. Both groups recovered their caches four hours after their second caching phase. Before the experiment, the birds had learned that worms would degrade within 120 hours, making them undesirable to recover. In the experiment, the birds that cached worms second would preferentially recover the worms because they had only been buried for four hours and were still fresh. The significant finding of this experiment was that the birds that had cached the worms first would recover the peanuts instead, implying that they could remember that the older worms were rotten and not to recover them.

This experiment showed that Western scrub jays can encode what-where-when information into memories. *What* because they also remembered qualitative information (whether the individual cache had worms or peanuts). *Where* because they were able to correctly remember what locations in the ice cube tray they had stored food. And *when* because they had the ability to incorporate information about when they cached each food and use it to inform their decision about which food to recover, causing them to avoid the worms if a longer time had passed. Overall, they have a representation of the kind of food and its location, and in some way can represent elapsed time. Because of this, it seems like birds might have a similar kind of memory to human episodic memory.

The Bischof-Köhler hypothesis (BKH) is similar to the MTT hypothesis, but it states that nonhuman animals don't have the ability to plan for future events. Since it doesn't involve consciousness like the MTT hypothesis, it's a lot easier to put to the test. An experiment by Raby et al. (2007) again used Western scrub jays to test whether they could anticipate their future needs while caching food. Across several days, the jays were either put into a

“no-breakfast” compartment, where they didn’t eat for two hours in the morning, or were put into the “breakfast” compartment, where they had access to pine nuts. In the evening, they were not given food for a while so that they were hungry. The birds were then given pine nuts that they could choose to eat immediately or to cache in the “breakfast” or “no-breakfast” compartment. The jays tended to choose to cache the majority of the pine nuts in the “no-breakfast” compartment, indicating that they were choosing to ignore their current hunger and instead planning for their future hunger the next morning. This finding challenges the Bischof-Köhler hypothesis by showing that birds can plan for future motivational states and ignore their current needs.

Although it’s impossible to know whether birds are actually projecting themselves forward and backward in time during these experiments, like the MTT hypothesis says, it’s definitely clear that they have the capacity to encode complex memories and predict future needs. Additionally, they can use these cognitive abilities to make decisions in the present. How do their brain structures enable this kind of cognition? Different mammals that can also do these tasks have similar brain regions that connect the hippocampus to the frontal cortex and help integrate what-where-when information. Even though birds’ brains have a very different overall organization, they seem to share this homologous network (Carruthers, 2013). This could mean that the way they remember and plan broadly involves the same algorithms or mechanisms as primates.

Birds that cache have different limits to the amount of spatial memory that they can store. For example, birds that live in environments that don’t have much food available, like Clark’s nutcrackers that live in the tundra, have a very high capacity to remember large numbers of cache sites. On the other hand, it’s possible that other caching birds might have adapted their spatial memory abilities to encode where-what-when memories about specific caches rather than high numbers of cache sites (Emery 2016). One trend across birds that cache, though, is that their hippocampus tends to be relatively larger than non-caching birds. Interestingly, the size of the hippocampus is dynamic and seems to change depending on their food-storing needs during the course of the year (Barnea & Pravosudov, 2011). The brains of adult birds broadly show high rates of neuron birth, or neurogenesis, across the brain but also specifically in the hippocampus. Mammals tend to have more specific areas of adult neurogenesis, which includes the hippocampus. What does this mean for learning and memory? Some research has found that higher rates of hippocampal neurogenesis correlate with better performance in learning and memory tasks in rats (Braun & Jessberger, 2014). Given how bird hippocampi also seem to correlate to memory needs, it seems that neuron birth might enable memory formation. But we still don’t know if that’s the case, or what widespread adult neurogenesis might mean in other parts of the avian brain.

### Section: The Evolution of Language

Birds can communicate in a lot of different ways, and the complexity of communication varies across species. All birds can produce calls, but only certain species can produce songs. Calls can be between members of the same species or different ones, and can communicate anything from a nearby predator to helping to keep members of a flock in touch. Bird calls aren’t learned and tend to be simple in structure, but can be used to communicate diverse kinds of

information. Songs, on the other hand, are more complex and must be learned. About half of all birds are oscines, or songbirds. In addition to songbirds, parrots and hummingbirds can also learn complex vocalizations. Interestingly, this ability is thought to have evolved independently in these three different lineages, showing that different evolutionary events can produce remarkably similar behaviors (Deacon, 2007).

Many aspects of learned bird vocalizations seem to be similar to human languages, although lacking the same level of complexity. One way that songbirds are similar to humans is in the learning process for vocalizations. There are two phases of song learning: the sensory acquisition phase and the sensorimotor phase (Emery, 2016). In the sensory acquisition phase, birds listen to songs sung by a tutor, normally their fathers. They can produce innate songs that are more simple in this phase, but aren't able to produce complex songs yet. This is followed by a silent period, where birds don't produce their own vocalizations. In the sensorimotor phase, birds will begin to integrate learned songs into their preexisting templates until they are able to directly copy the learned song. Like humans, some species of birds have a sensitive period for learning (Mooney, 1999). If they don't hear songs in this period, they can't produce complex songs as adults, although they still can produce some kinds of vocalizations. Unlike humans, other species of birds are not age-limited learners and can continue to learn new vocalizations throughout their lives (Emery, 2016). These bird species are particularly adept at imitation, which is a capacity shared by very few other animals.

Marc D. Hauser, an evolutionary biologist, argues that some faculties of language exist in a broad sense and aren't uniquely human (Houser et al., 2002). Vocal imitation, in particular, is a prerequisite for the faculty of language in a broad sense, since it's a necessary component for language learning and communication. Interestingly, primates and monkeys don't have an extensive ability to imitate, implying that this was a recent evolutionary development that might have arisen in humans (Houser et al., 2002). Unlike primates, birds do have an extensive imitative capacity. Why? In one capacity, it enables birds to learn songs. In non-songbirds, like parrots, it's less clear why having the ability to imitate vocalizations would be advantageous. Some birds imitate predators to scare off their rivals, or imitate their social partner as a way to bond (Emery, 2016). So, imitation is a shared ability of humans and birds, although it seems to play slightly different roles across species.

Whether birds learn referential communication (transmitting information through sound about a specific referent) is a topic of debate. One important source of information about this topic are experiments with Alex, an African grey parrot. Alex was involved in studies for 30 years and gave scientists many important insights into avian cognition. He was trained to learn labels and concepts for over 50 objects. Over time, he learned how to answer questions about the color, shape, and material of these objects, as well as being able to integrate conceptual information. For instance, he could correctly complete tasks like naming the number three-cornered objects in a collection of objects. He could also accomplish more conceptually complicated tasks, like naming which objects in an array were the same or different (Pepperberg, 1987). Beyond that, Alex also had the ability to count sets of up to eight objects and sum two sets of up to six objects (Pepperberg, 2012). All of these tasks indicate that he had comparable cognitive abilities to non-human primates.

Even though the experiments with Alex were a powerful example of how birds can learn complicated communication, it would be a stretch to say that he was able to use language. Alex could engage in symbolic communication by associating symbols with meaning (semantics), but he couldn't express himself beyond this level. Birdsong, in contrast, doesn't have semantics, words, or syntax. The similarity between birdsong and human language is that both consist of "complex, patterned vocalizations" (Berwick et al., 2011). Even the most complex bird songs can be described with a set of probabilistic rules between a finite number of states (Berwick et al., 2011). So, songbirds can be a useful model to study the underlying neural and learning control of language in humans, but since they lack semantics and syntax, they aren't necessarily a helpful model for understanding how these elements exist in humans.

How are language abilities encoded in the brain? Even though the structures of brains across hummingbirds, parrots, and songbirds are all different, they seem to have functionally equivalent regions that are responsible for vocalizations. Each has circuitry that connects auditory inputs (things they hear) to motor outputs (making vocalizations) through regions of the forebrain. Listening to and producing songs requires the coordination of about a dozen areas in the forebrain (Deacon, 2007). Birds that can't learn vocalizations don't have these complex circuits. In songbirds, the higher vocal center (HVC) seems to be central to controlling song flexibility and complexity. A lot like the hippocampus, new neurons are born at high rates during breeding season, when singing capacity is especially important (Emery, 2016). Parrots also have some notable differences in brain structure compared to songbirds. They have a different motor output pathway, and a direct pathway between their forebrain projections to motor output systems (whereas songbirds have regions of the midbrain that mediate these connections) (Deacon, 2007). Understanding more how variations in vocalizations exist between the brains of different birds could give us useful insights to understanding how language in humans evolved.

### Section: Social intelligence

Most birds live together in groups and therefore need to have certain social cognitive abilities. Some evolutionary psychologists support a "social intelligence hypothesis," which postulates "that cognitive capacities are most likely an adaptation to life in complex social groups" (Heyes, 2012; Kaminski, 2017). These cognitive capacities arose to help individuals to flexibly respond to changing social dynamics. Another hypothesis, the "Machiavellian Intelligence hypothesis," argues that complex cognition is tied to competition within social groups, and that having the ability to outwit others can give individuals a competitive advantage (Kaminski, 2017). Both hypotheses argue that social living helped drive complex social cognition in humans, which would imply that birds might have social cognitive skills that can be compared to humans. For birds, living in larger groups can make it easier to find a mate, raise offspring, and collectively avoid predators (Emery, 2016). Social intelligence helps birds to understand social hierarchies, learn from others, and cooperate to solve problems.

Birds that live in large social groups will create complex social hierarchies, which is important for avoiding conflict between conspecifics (members of the same species). A certain cognitive skill set is necessary to be able to form these kinds of hierarchies. One skill that is particularly important is having the ability to recognize other individuals and remember their place in the social hierarchy over long periods of time. Because social groups are dynamic and

constantly undergo changes in membership, it's also useful to be able to extrapolate the social position of an unknown individual. This ability requires "transitive inference," which is a kind of deductive reasoning (Emery, 2016). A study with pinyon jays, a particularly social species of corvid, found that they can use transitive inferences to inform their behavior (Paz-y-Miño et al., 2004). In the experiment, birds were split into three different social groups and independently formed group social hierarchies. Later, jays would observe an unknown individual from a different group interact with a known individual from their group. When the jay had to later interact with the unknown individual, it behaved appropriately with respect to its social rank, even though it wasn't aware of the overall hierarchy across the three groups. This is an exciting finding because transitive inference, which is a skill that develops in humans at around five years of age, was thought to be rare in non-human animals (Paz-y-Miño et al., 2004). It also indicates that evolutionary pressures toward sociality can lead to complex reasoning abilities.

Another implication of social living is the ability to cooperatively solve problems with members of one's social group. In an experiment by Seed et al. (2008), they found apparent limitations in the ability for corvids to flexibly solve problems together. Their experiment involved two rooks having to work together to pull both ends of a string to make it taught. When the string was taught, it would push bowls closer to the rooks and allow them to access the food inside. If the rook worked alone at this task, it would pull the string out of place and make it impossible to access the food. The experimenters found that each rook could learn to cooperate with a social partner and solve the task. But did these birds understand that they needed a partner in order to complete the task? Studies with primates seem to indicate that they understand the need for cooperation in this task (Bugnyar, 2008). Unlike primates, rooks didn't take advantage of their partner if they were given the opportunity to monopolize their access to food during the task, indicating they might not have an understanding of how the task worked. Also, if there was a delay for the partner to enter the room, the first rook would attempt to complete the task by itself, which chimps didn't do in this same experimental set-up. This is, of course, a single study, but it might imply that corvids might not actually understand the requirements of the task in the same way that primates do.

### Section: Theory of Mind

Theory of mind (ToM) is the ability to attribute mental states to oneself and others. Specifically, ToM is thought to require the formation of "meta-representations," which are representations that are decoupled from reality (Frith & Frith, 2005). The concept of "Theory of Mind" in non-human animals is controversial, to say the least. Are non-human animals conscious? Do they have a sense of self? Without language, it's difficult to differentiate associative learning from theory of mind. Are birds just learning to associate certain behaviors with outcomes, or are they actually able to attribute mental states to their conspecifics?

It does seem that corvids can attribute desires to their conspecifics. Many species of birds form life-long partnerships and engage in many specific social behaviors with their partners. Ostojic et al. (2013) conducted a study in pairs of Eurasian jays, a kind of corvid, to test whether males could attribute desires to their partners. Males often socially share food with their partners, which was central to this experiment. Additionally, corvids, like humans, exhibit "satiety," which is a decreased desire for a specific food after eating a large amount of it. In the

experiment, male jays either observed, or didn't observe, their female partner eating a specific food to satiety. The males were then given a choice in the type of food to share with their partner. The males that hadn't observed their partners eating tended to choose the food randomly, but the males that had observed their partners eating a specific food tended to bring her the other kind of food. This indicates that the jays understood that their past experiences with satiety might also apply to their female partner, causing her to prefer to eat new kinds of food rather than the same kind. The results of this study indicate that corvids can possibly attribute desire, which is thought to be an evolutionary precursor of ToM.

Some evidence shows that birds might be able to project their experiences to their conspecifics, which could imply a ToM. While caching, Western scrub jays must be careful to remember where they hid their food, but also have to prevent thieving jays from stealing their cache. In order to avoid having their cache pilfered by their conspecifics, jays engage in a bunch of counter-strategies to effectively hide their cache. One common strategy is to re-cache food out of the sight of potential thieves. Dally et al. (2010) found that scrub jays would re-cache food if they were observed during the original caching, but they would only do so if they themselves had experience with stealing food. If the scrub jay didn't have direct experience with pilfering the caches of others, it didn't re-cache. These results are interesting because they could indicate that these scrub jays projected their past pilfering experience onto their conspecifics, and used it to inform their decisions to re-cache if they were observed.

So, do birds form meta-representations? It's possible that animals in the lab have been conditioned to make certain associations rather than actually attributing mental states to others. Another possible explanation is that animals can attribute behavior to their conspecifics, but not mental states. Even so, these kinds of abilities have only been found in corvids and primates, implying that there has been convergent evolution to produce similar kinds of complex cognition in both kinds of animals.

## Conclusions

Because avian cognition and intelligence is a relatively new area of neuroscience, there's still a lot of work to be done to understand how they perform in certain cognitive tasks compared to humans and non-human primates. Importantly, though, "evidence of behavioral convergent evolution in corvids and apes is not evidence of cognitive convergent evolution" (Taylor, 2014). Although there seems to be many similarities in how some bird species and primates behave, it's not clear how cognition is carried out in the brain.

What is the best way to understand human cognition and intelligence? What level of functionality matters? And to what extent can we say that human and avian brains are similar? The philosopher Hilary Putnam's concept of 'functional isomorphism' can be useful to analyze these two systems. Two systems are functionally isomorphic if there is "a correspondence between the states of one and the states of the other that preserves functional relations" (Putnam, 1975). Essentially, he argues that the constituents of a system don't have to be the same in order to consider them functionally equivalent. Although avian and mammalian brains are structured very differently, the broad way that regions connect seem to more or less be equivalent (Shanahan et al., 2013). So, can we say that avian brains are functionally isomorphic

to human brains? And can this approach help us make general conclusions about how intelligence might exist in the brain?

Evolutionary pressures in birds and humans seem to have produced some similar cognitive capacities, especially when it comes to social cognition. Studying avian cognition might even be better in some cases than primate cognition, since certain species of birds can outperform non-human primates in some cognitive tests. In particular, studying songbirds and parrots could be particularly useful for drawing conclusions about how human language abilities might be elaborated in the brain.

Errors in cognition can also be a useful way to understand how different brains work. Cognitive tests in the lab have shown that birds make certain mistakes that primates don't make. For instance, corvids make specific kinds of mistakes in causal understanding during tool use studies (Taylor, 2014). They also make certain unique mistakes during cooperative problem solving, as discussed earlier. Does that mean that they use different algorithms to solve certain problems than primates do? It can be difficult to tell based only on studying behavioral outputs.

There are of course limitations to a comparative cognition approach. Computer scientist Allen Newell once criticized the experimental methods in cognitive psychology, stating, "you can't play 20 questions with nature and win" (Kriegeskorte & Douglas, 2018). Scientists have to create tasks that test a single cognitive hypothesis at a time, which can make it difficult to come to more generalizable conclusions about avian intelligence. Kriegeskorte and Douglas (2018) argue that human cognition can be best understood through neurobiologically plausible artificial intelligence models. I would argue that, while using AI can be an important approach to understanding cognition, comparative cognition studies also provide essential information on how intelligence evolved through evolutionary constraints. Starting to examine more of the underlying neurobiology associated with avian cognition can give us important insights into our own intelligence and cognition.

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